EXP.NO: 6

AIM:

To demonstrate session tracking using HttpSession in a Servlet-based Personal Portfolio Builder application by implementing a simple login system where the user's session is maintained.

ALGORITHM:

1. Input:

- **The user enters username and password in the login form.**
- **The servlet checks the credentials and creates a session if valid.**

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2. Steps for Implementing Session Tracking using HttpSession:

1. Create the Login Form (HTML):

- **Design an HTML page with input fields for username and password.**
- **Add a submit button to send data to the Login Servlet using POST method.**

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2. Create the Login Servlet:

- **Develop a servlet to handle login requests.**
- **Retrieve username and password from the request.**
- **Validate the credentials (hardcoded values for simplicity).**
- **If valid, create a new session using HttpSession.**
- **Store the username in the session using setAttribute().**

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3. Validate User Credentials:

- **Compare entered username and password with predefined values.**
- **If valid → create session and redirect to portfolio page.**
- **If invalid → display error message and reload login page.**

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4. Create and Track User Session:

- **Use HttpSession object to maintain user session.**
- **Store user data like username in session.**
- **Session helps to track the user across pages.**

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5. Redirect to Portfolio Page:

- **After successful login, redirect the user to the Portfolio Dashboard page.**
- **Check session in the dashboard page before displaying content.**
- **If session is not available, redirect back to login page.**

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6. Logout and Session Invalidation:

- **Provide a logout option.**
- **When user clicks logout, invalidate session using session.invalidate().**
- **Redirect user back to login page.**

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7. Session Expiration:

- **If session expires or user logs out, session data is removed.**
- **User must login again to access the portfolio page.**

Program

Login.html

HTML

```
<!DOCTYPE html>
<html>
<head>
    <title>Personal Portfolio Login</title>
    <style>
        body {
            font-family: Arial;
            background: linear-gradient(to right, #667eea, #764ba2);
            display: flex;
            justify-content: center;
            align-items: center;
            height: 100vh;
        }

        .login-box {
            background: white;
            padding: 30px;
```

```
            border-radius: 10px;
            box-shadow: 0 0 15px rgba(0,0,0,0.2);
            width: 300px;
            text-align: center;
        }

        input {
            width: 90%;
            padding: 10px;
            margin: 10px 0;
            border-radius: 5px;
            border: 1px solid #ccc;
        }

        button {
            background: #667eea;
            color: white;
            border: none;
            padding: 10px;
            width: 100%;
            border-radius: 5px;
            cursor: pointer;
        }

        button:hover {
            background: #5a67d8;
        }
    </style>
</head>

<body>

<div class="login-box">
    <h2>Portfolio Login</h2>

    <form action="LoginServlet" method="post">
        <input type="text" name="username" placeholder="Enter Username" required>
        <input type="password" name="password" placeholder="Enter Password"
required>
        <button type="submit">Login</button>
    </form>
</div>
```

```
</body>
</html>
```

LoginServlet.java

Java

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.*;

@WebServlet("/LoginServlet")
public class LoginServlet extends HttpServlet {

    protected void doPost(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {

        String username = request.getParameter("username");
        String password = request.getParameter("password");

        response.setContentType("text/html");

        // Simple authentication (hardcoded)
        if (username.equals("admin") && password.equals("1234")) {

            HttpSession session = request.getSession();
            session.setAttribute("user", username);

            response.sendRedirect("PortfolioServlet");

        } else {
            PrintWriter out = response.getWriter();
            out.println("<h3>Invalid Username or Password</h3>");
            out.println("<a href='Login.html'>Try Again</a>");
        }
    }
}
```

PortfolioServlet.java (Your Dashboard Page)

Java

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.*;

@WebServlet("/PortfolioServlet")
public class PortfolioServlet extends HttpServlet {

    protected void doGet(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        HttpSession session = request.getSession(false);

        if (session != null && session.getAttribute("user") != null) {

            String user = (String) session.getAttribute("user");

            out.println("<html><head><style>");
            out.println("body { font-family: Arial; background:#f4f6f9; padding:20px; }");
            out.println(".header { background:#333; color:white; padding:15px; border-
radius:8px; }");
            out.println(".box { background:white; margin:15px 0; padding:15px; border-
radius:8px; box-shadow:0 0 10px rgba(0,0,0,0.1);} ");
            out.println(".logout { display:inline-block; margin-top:20px; padding:10px;
background:red; color:white; text-decoration:none; border-radius:5px; }");
            out.println("</style></head><body>");

            out.println("<div class='header'><h2>Personal Portfolio Builder</h2></div>");
            out.println("<h3>Welcome " + user + "</h3>");

            out.println("<div class='box'>You can now create your portfolio
website.</div>");
            out.println("<div class='box'>Add your details, projects and skills.</div>");
            out.println("<div class='box'>Upload your resume and share your GitHub
profile.</div>");

            out.println("<a class='logout' href='LogoutServlet'>Logout</a>");

            out.println("</body></html>");
```

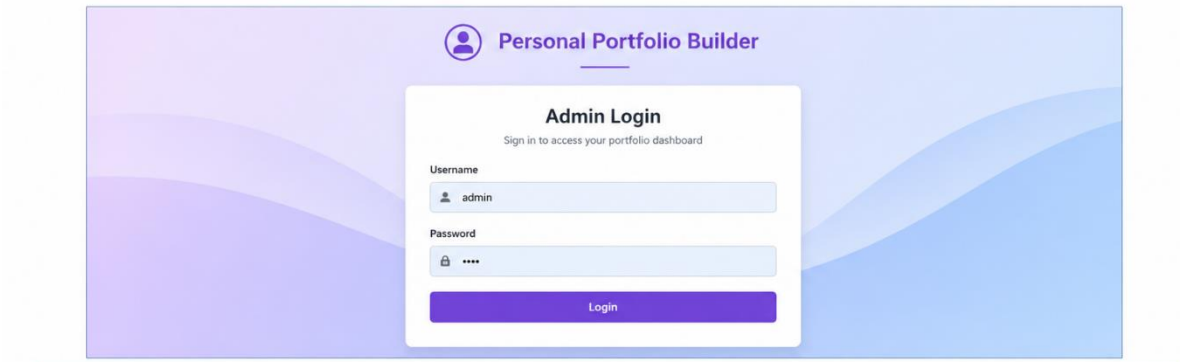
```
        } else {
            response.sendRedirect("Login.html");
        }
    }
}
```

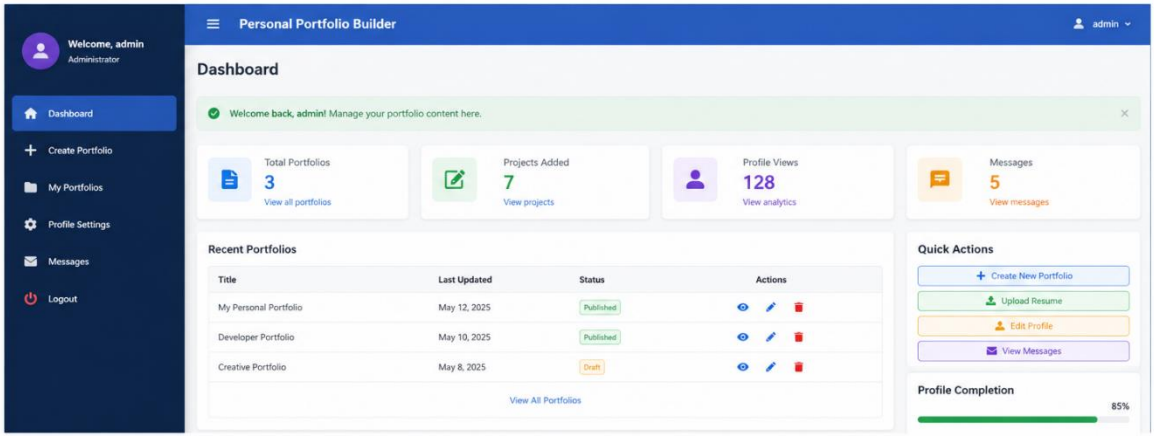
LogoutServlet.java

Java

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.*;

@WebServlet("/LogoutServlet")
public class LogoutServlet extends HttpServlet {

    protected void doGet(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {

        HttpSession session = request.getSession(false);

        if (session != null) {
            session.invalidate();
        }

        response.sendRedirect("Login.html");
    }
}
```

Output:





Result:

Session tracking using HttpSession is successfully implemented in the Personal Portfolio Builder project. User login session is created, maintained, and destroyed using servlets.